Syllabus Formalized Mathematics in Lean
(Computer-assisted Mathematics – V3A6 / F4A1) · WiSe 26/27

Instructors:
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Class times
Lectures on Wednesday 16 – 18 and Friday 14 – 16 in the KHS
Exercise sessions are

- Tuesday 12 – 14 in SR 0.006. Tutor: **Abel Doñate**, s63adona@uni-bonn.de
- Friday 8 – 10 in SR 0.006. Tutor: **Felix Pernegger**, s96pcage@uni-bonn.de

Office Hours
There is a Lean hacking session every Tuesday 18:00 – 20:00 in seminar room 0.011, which you can join to work on Lean exercises, projects or to ask questions. A personal meeting can be scheduled upon request.

About the course
In this course you will learn how to explain mathematical theories to a computer using a computer program called Lean. Using the language of Lean you can write mathematical definitions, theorems and proofs, and then Lean can check whether your proofs are correct and contain no holes. In this course we will learn how to interact with Lean and write your own proofs in it, and we will prove basic results in various mathematical topics, including algebra, topology and analysis. The course involves weekly homework exercises, a small and a large project, and a final written exam.

Goals

- Learn to explain mathematics to the computer program Lean
- Improve your understanding of mathematical proofs
- Learn some new mathematics
- Contribute to projects using Git
- Learn computer skills, such as using VSCode
- Personal project management skills
- Responsible use of AI in mathematics

Resources

- The online textbook *Mathematics in Lean* by Jeremy Avigad and Patrick Massot
- Learning resources on the Lean website:
  https://leanprover-community.github.io/learn.html

Registration
Make sure to register for the course on eCampus. Register for the exam on Basis.

Homework
For a large part of the course, there will be a weekly set of homework problems assigned on Wednesday. You are strongly encouraged to hand-in the homework, so that a tutor will grade it, but it is not mandatory. Homework is due one week later on Wednesday before class.

Projects
You will be required to complete two projects and hold one presentation as part of the course, all three of which must be satisfactory to be admitted to the final exam.
The first project consists of contributing to an open source project, by formalizing the statement of an open conjecture in mathematics.
The second project consists of proving a small mathematical theorem or developing a bit of theory by yourself or with one partner.
You also have to present your project during the exercise classes in a (roughly) 15 minute presentation.
More information about the projects will be given during the lectures. More information about the presentations will be provided by the tutor.

Generative AI Policy
Generative AI is very good at writing Lean, and can solve all exercises in this class. However, if you just ask AI to solve the exercises for you, you will not learn anything. Therefore, any exercise or project that you hand in should be written by you.
Allowed ways of using generative AI is to use it like a tutor. Asking questions about a Lean concept or tactic is fine. Asking for a hint is fine. Asking it to generate a complete proof for you is not.
Depending on the severity, violations can result in a warning or refusal to admit you to the exam.

Final Exam
There will be a written final exam, which will be *closed-book and no laptop/electronics allowed.* This will test whether you know what you're doing when using Lean. No knowledge of nonstandard tactics or Mathlib declarations will be required.
To be admitted to the final exam, your projects must be satisfactory.
The final exam is in the last week of classes (during class time on Wednesday). There will (obviously) be no regular class that week.

First exam time: Wed 03.02.2027 at 14 – 16 (GHS)
Second exam time: Thu 28.03.2027 at 9 – 11 (GHS)

Grade: Your final grade is fully determined by the final exam grade.

Code of conduct and ombudspersons:
Everyone attending this class will contribute to an inclusive and welcoming environment where we all treat each other professionally and with mutual respect, regardless of origin, beliefs, physical ability, gender or sexual identity. Discriminatory, racist, sexist, exclusionary, bullying, or harassing behavior will not be tolerated. In the event that you witness inappropriate behavior, please consider

intervening if it is safe for you to do so and / or informing an ombudsperson. You might also consider contacting the victim and offering help.
The ombudspersons can be contacted at any time in the event of conflict between individuals, perceived misconduct, or any form of harassment. They are bound by secrecy.
The ombudspersons in mathematics are currently:
Dr. Regula Krapf (krapf@math.uni-bonn.de), Mathematical Institute
Dr. Richard Schubert (schubert@iam.uni-bonn.de), Institute for Applied Mathematics
For more information, see
https://www.mathematics.uni-bonn.de/en/department/fachgruppe-mathematik#fgombud